# Module 1: Introduction to Relational Databases

COMP 2714 – Relational Database Systems

Learning Outcome: LO1 – Describe the functions and characteristics of the relational database model

# What you will learn …

At the end of this class you will be able to:

- Define the **difference between data and information**
- See how modern databases evolved from **file systems**
- Understand **flaws** in file system data management
- Describe what a **database** is, various types of database, and why they are valuable assets for decision making
- Explain the **importance of database design**
- Outline the **main components** of the database system
- Describe the main functions of a database management system (**DBMS**)

# Everybody is talking about Data!

- Why is it that **data & information** have become so important?
- Data is nothing extraordinary - everything that captures our attention and is **relevant** to us to know is data.
- We, as humans have to have information **to make decisions.**

**But what is information and what do we exactly mean by data?**

# Data & Information

- Data are the **raw facts**
- Information is the result of processing raw data to reveal its **meaning**
- To produce information you need to start with a **question**

How do we **store, process and manage** data and information?

# Why Databases?

- We need information **to make better decisions**
- Our brain has evolved to **create categories of information** and then somehow **link them together**, building a structure
- Our memory accesses the data & information we need through these structures
- With **Relational Database Design**, we are trying to do (for the computer) what our brain does naturally: defining structures for storing data to make *finding*, *changing* and *managing* information more seamless, efficient and reliable.

Databases are **the tool** for data & information storage and management. They have **evolved over decades** to respond to our **information needs**.

# Evolution of File System Data Processing

1. Manual file systems

   Accomplished through a system of **file folders and filing cabinets**

2. Computerized file systems

   **Data processing (DP) specialist** created a computer-based system to track data and produce required reports

3. Modern end-user productivity tools (kind of going back to file systems! )

   Includes **spreadsheet programs** such as Microsoft Excel

# Problems with File System Data Processing

- Lengthy development times
- Difficulty of getting quick answers
- Complex system administration
- Lack of security and limited data sharing
- Extensive programming

# Structural and Data Dependence (1 of 2)

**Structural dependence**

- Access to a file is **dependent on its own structure**
- All file system programs are modified to conform to a new file structure

Structural **independence**

- File **structure is changed without affecting** the application's ability to access the data

# Structural and Data Dependence (2 of 2)

**Data dependence**

- Data access changes when **data storage characteristics** change

Data **independence**

- Data storage characteristics are changed **without affecting** the program's ability to access the data

Practical **significance** of data dependence is the difference between **logical and physical format**

# Logical & Physical Data Format

- Logical data format is how **humans** view the data
- Physical data format is how the **computer** must work with the data
- Data dependence makes managing the file system extremely **cumbersome**
- Every file access program must tell the computer not only **what to do** but also **how to do it**

# Data Redundancy (1 of 2)

- It is **difficult to combine data** from different sources in a file system environment
- Often different users/ departments keep their own copy of the data (**islands of information**)
- these data are **not consistently updated** and each contain a slightly different version of the same data
- **Data Redundancy:** when the same data is unnecessarily kept at **different locations**

# Data Redundancy (2 of 2)

Possible results of **uncontrolled data redundancy**

- Poor data security
- Data inconsistency
- Data-entry errors
- Data integrity problems

# Data Integrity

Data Integrity is present when the **data corresponds to reality.** that the data is **accurate** *and* **consistent**. data integrity is closely related to data consistency.

# Data Anomalies

Develops when not all of the **required changes** in the redundant data are made successfully. e.g:

- Update anomalies
- Insertion anomalies
- Deletion anomalies

# Database Systems

- the file system consists of **many separate and unrelated files**
- appears to the end user, as **a single logical unit**
- Enables end user to only think of **what to do** and **not how to do it**.
- DBMS eliminates data inconsistencies, data anomaly, data dependence and structural dependence problems, *if* the database is **designed properly**.
- **DBMS** stores the data structures, relationships between structures, and access paths

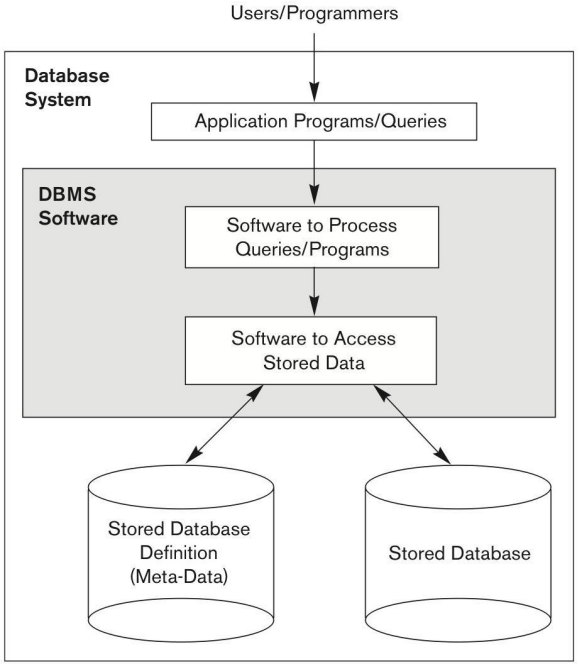
# Database Defined

**Database:**

- A shared, integrated computer structure that houses a collection of related data.
- contains two types of data: **end-user data** (raw facts) and **metadata.**

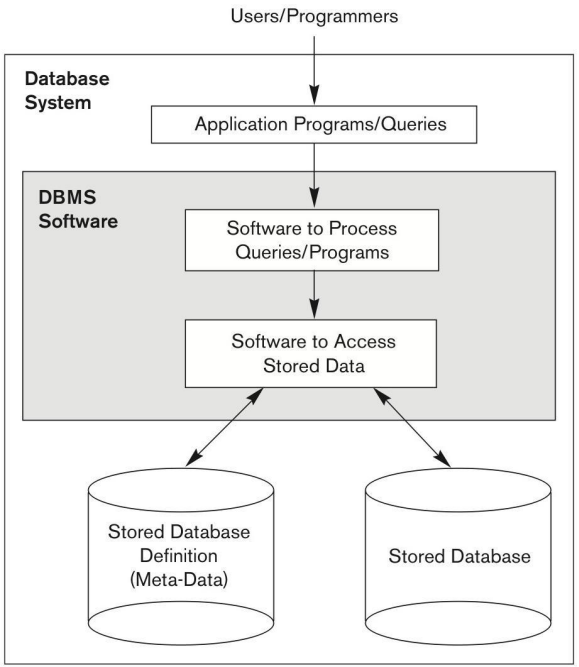
**Metadata:**

- Data about data; that is, data about data characteristics and relationships.



# DBMS Defined

- collection of programs that manages the database structure and controls access to the data stored in the database.
- acts as an intermediary between the end use and the database hiding much of the database internal complexity and presents the use with a single, integrated view of the data.
- Access to the database is only through the DBMS
- makes it possible to share data across multiple applications



# DBMS Functions (1 of 3)

## Data dictionary management

Data dictionary: stores definitions of data elements and their relationships

## Data storage management

Performance tuning ensures efficient performance

## Data transformation and presentation

Data is formatted to conform to logical expectations

# DBMS Functions (2 of 3)

## Security management

Enforces user security and data privacy

## Multi User access control

Sophisticated algorithms ensure that multiple users can access the database concurrently without compromising its integrity

## Backup and recovery management

Enables recovery of the database after a failure

# DBMS Functions (3 of 3)

## Data integrity management

Minimizes redundancy and maximizes consistency

## Database access languages and application programming interfaces

- Query language: lets the user specify what must be done without having to specify how
- Structured Query Language (SQL): de facto query language and data access standard supported by the majority of DBMS vendors

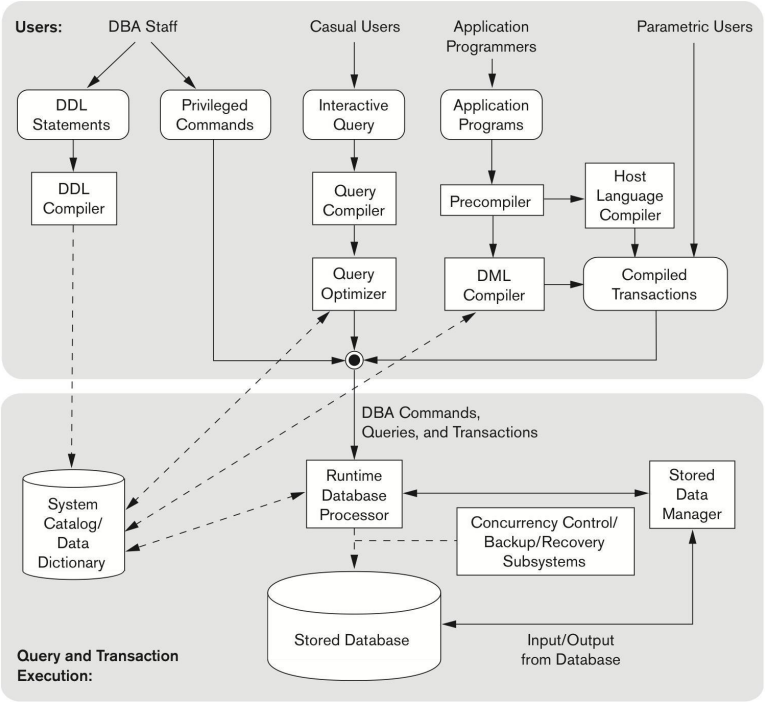
## Database communication interfaces

Accept end-user requests via multiple, different network environments

# The Database System Environment

Database system: organization of **components** that define and regulate the collection, storage, management, and use of data within a database environment

- Hardware
- Software
- People
- Procedures
- Data



# Mini-world Example

**STUDENT**

| Name | Student_number | Class | Major |
|---|---|---|---|
| Smith | 17 | 1 | CS |
| Brown | 8 | 2 | CS |

**COURSE**

| Course_name | Course_number | Credit_hours | Department |
|---|---|---|---|
| Intro to Computer Science | CS1310 | 4 | CS |
| Data Structures | CS3320 | 4 | CS |
| Discrete Mathematics | MATH2410 | 3 | MATH |
| Database | CS3380 | 3 | CS |

**SECTION**

| Section_identifier | Course_number | Semester | Year | Instructor |
|---|---|---|---|---|
| 85 | MATH2410 | Fall | 07 | King |
| 92 | CS1310 | Fall | 07 | Anderson |
| 102 | CS3320 | Spring | 08 | Knuth |
| 112 | MATH2410 | Fall | 08 | Chang |
| 119 | CS1310 | Fall | 08 | Anderson |
| 135 | CS3380 | Fall | 08 | Stone |

**GRADE_REPORT**

| Student_number | Section_identifier | Grade |
|---|---|---|
| 17 | 112 | B |
| 17 | 119 | C |
| 8 | 85 | A |
| 8 | 92 | A |
| 8 | 102 | B |
| 8 | 135 | A |

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---|---|
| CS3380 | CS3320 |
| CS3380 | MATH2410 |
| CS3320 | CS1310 |

# Metadata & Catalog

- Metadata = data about data
- Catalog stores schema, constraints, data types

**RELATIONS**

| Relation_name | No_of_columns |
|---|---|
| STUDENT | 4 |
| COURSE | 4 |
| SECTION | 5 |
| GRADE_REPORT | 3 |
| PREREQUISITE | 2 |

**COLUMNS**

| Column_name | Data_type | Belongs_to_relation |
|---|---|---|
| Name | Character (30) | STUDENT |
| Student_number | Character (4) | STUDENT |
| Class | Integer (1) | STUDENT |
| Major | Major_type | STUDENT |
| Course_name | Character (10) | COURSE |
| Course_number | XXXXNNNN | COURSE |
| …. | …. | ….. |
| …. | …. | ….. |
| …. | …. | ….. |
| Prerequisite_number | XXXXNNNN | PREREQUISITE |

# Database Users – Actors on the Scene

- DBAs: manage system, authorize access
- Designers: define structure and constraints
- End users: casual, naive, sophisticated, standalone

# Database Users – Behind the Scene

- System implementors
- Tool developers
- Operators/maintenance staff

# Why Database Design is Important

- People naturally have a **“narrow” view** of the data in their environment.
- this narrow view and **lack of proper data modelling** and database design skills leads to poor database design choices which is **typical in spreadsheet** and and desktop database programs (e.g. MS Access)
- **Database Design**, is to build the structure that facilitates storing and managing end-user data. So it meets all user requirements.
- Even a good DBMS performs poorly with a badly designed database
- To properly design a database we must first **precisely identify databases expected use**

## FIGURE 1.5 EMPLOYEE SKILLS CERTIFICATION IN A POOR DESIGN

Why are there blanks in rows 9 and 10?

How to produce an alphabetical listing of employees?

How to count how many employees are certified in Basic Database Manipulation?

Is Basic Database Manipulation the same as Basic DB Manipulation?

What if an employee acquires a fourth certification?

Do we add another column?

| ID | ENum | Name | Title | HireDate | Skill1 | Skill1Date | Skill2 | Skill2Date | Skill3 | Skill3Date |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 02345 | Brian Oates | DBA | 2/14/1997 | Basic Database Management | 2/14/2004 | Advanced Database Management | 2/14/2007 | Basic Web Design | 8/9/2005 |
| 2 | 08273 | Marco Bienz | Analyst | 7/28/2008 | Basic Web Design | 3/8/2011 | Advance Process Modeling | 8/19/2014 | | |
| 3 | 06234 | Jasmine Patel | Programmer | 8/10/2007 | Basic Web Design | 8/10/2009 | Advanced C# programming | 8/10/2009 | Basic DB manipulation | 1/29/2014 |
| 4 | 03373 | Franklin Johnson, Jr. | Purchasing Agent | 3/15/2004 | Advanced Spreadsheets | 6/20/2013 | | | | |
| 5 | 13567 | Almond, Robert | Analyst | 9/30/2014 | Basic Process Modeling | 9/30/2016 | Basic Database Design | 5/23/2017 | | |
| 6 | 10282 | Richardson, Amanda | Clerk | 4/11/2013 | | | | | | |
| 7 | 09382 | Susan Mathis | Database Programmer | 8/2/2012 | Basic DB Design | 8/2/2014 | Basic Database Manipulation | 8/2/2014 | Advanced DB Manipulation | 5/1/2015 |
| 8 | 14311 | Duong, Lee | Programmer | 9/1/2016 | Basic Web Design | 9/1/2018 | | | | |
| 9 | | | | | Master Database Programming | | | | | |
| 10 | | | | | Basic Spreadsheets | | | | | |
| 11 | 09002 | Wade Gaither | Clerk | 5/20/2012 | Advanced Spreadsheets | 5/16/2015 | Basic Web Design | 5/16/2015 | | |
| 12 | 13383 | Raymond F. Matthews | Programmer | 3/12/2014 | Basic C# Programming | 3/12/2016 | | | | |
| 13 | 09283 | Chavez, Juan | Clerk | 7/4/2012 | | | | | | |
| 14 | 04893 | Patricia Richards | DBA | 6/11/2006 | Advanced Database Management | 6/11/2008 | Advanced Database Manipulation | 9/20/2014 | | |
| 15 | 13932 | Lee, Megan | Programmer | 9/29/2015 | | | | | | |

## FIGURE 1.6 EMPLOYEE SKILLS CERTIFICATION IN A GOOD DESIGN

Table name: EMPLOYEE

| Employee_ID | Employee_FName | Employee_LName | Employee_HireDate | Employee_Title |
|---|---|---|---|---|
| 02345 | Brian | Oates | 2/14/1999 | DBA |
| 03373 | Franklin | Johnson | 3/15/2006 | Purchasing Agent |
| 04893 | Patricia | Richards | 6/11/2008 | DBA |
| 06234 | Jasmine | Patel | 8/10/2009 | Programmer |
| 08273 | Marco | Bienz | 7/28/2010 | Analyst |
| 09002 | Wade | Gaither | 5/20/2014 | Clerk |
| 09283 | Juan | Chavez | 7/4/2014 | Clerk |
| 09382 | Susan | Mathis | 8/2/2014 | Database Programmer |
| 10282 | Amanda | Richardson | 4/11/2015 | Clerk |
| 13383 | Raymond | Matthews | 3/12/2016 | Programmer |
| 13567 | Robert | Almond | 9/30/2016 | Analyst |
| 13932 | Megan | Lee | 9/29/2017 | Programmer |
| 14311 | Lee | Duong | 9/1/2018 | Programmer |

Database name: Ch01_Text

Table name: CERTIFIED

| Employee_ID | Skill_ID | Certified_Date |
|---|---|---|
| 02345 | 100 | 2/14/2004 |
| 02345 | 110 | 8/9/2005 |
| 02345 | 180 | 2/14/2007 |
| 03373 | 120 | 6/20/2013 |
| 04893 | 180 | 6/11/2008 |
| 04893 | 220 | 9/20/2014 |
| 06234 | 110 | 8/10/2009 |
| 06234 | 200 | 8/10/2009 |
| 06234 | 210 | 1/29/2014 |
| 08273 | 110 | 3/8/2011 |
| 08273 | 190 | 8/19/2014 |
| 09002 | 110 | 5/16/2015 |
| 09002 | 120 | 5/16/2015 |
| 09382 | 140 | 8/2/2014 |
| 09382 | 210 | 8/2/2014 |
| 09382 | 220 | 5/1/2015 |
| 13383 | 170 | 3/12/2016 |
| 13567 | 130 | 9/30/2016 |
| 13567 | 140 | 5/23/2017 |
| 14311 | 110 | 9/1/2018 |

Table name: SKILL

| Skill_ID | Skill_Name | Skill_Description |
|---|---|---|
| 100 | Basic Database Management | Create and manage database user accounts. |
| 110 | Basic Web Design | Create and maintain HTML and CSS documents. |
| 120 | Advanced Spreadsheets | Use of advanced functions, user-defined functions, and macroing. |
| 130 | Basic Process Modeling | Create core business process models using standard libraries. |
| 140 | Basic Database Design | Create simple data models. |
| 150 | Master Database Programming | Create integrated trigger and procedure packages for a distributed environment. |
| 160 | Basic Spreadsheets | Create single tab worksheets with basic formulas |
| 170 | Basic C# Programming | Create single-tier data aware modules. |
| 180 | Advanced Database Management | Manage Database Server Clusters. |
| 190 | Advance Process Modeling | Evaluate and Redesign cross-functional internal and external business processes. |
| 200 | Advanced C# Programming | Create multi-tier applications using multi-threading |
| 210 | Basic Database Manipulation | Create simple data retrieval and manipulation statements in SQL. |
| 220 | Advanced Database Manipulation | Use of advanced data manipulation methods for multi-table inserts, set operations, and correlated subqueries. |

Database design is a process of decomposition: the integrated data must be decomposed into its constituent parts. each part stored in its own table

# Relational Model Basics

- Relation (table) = set of tuples (rows) with attributes (columns)
- Tuple (row) = record of data
- Attribute (column) = property with a domain (data type)

**STUDENT**

| Name | Student_number | Class | Major |
|---|---|---|---|
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| Brown | 8 | 2 | CS |

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| 8 | 92 | A |
| 8 | 102 | B |
| 8 | 135 | A |

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---|---|
| CS3380 | CS3320 |
| CS3380 | MATH2410 |
| CS3320 | CS1310 |

# Keys in the Relational Model

- Primary key = unique identifier
- Candidate keys = alternative choices
- Foreign key = reference to another relation

**STUDENT**

| Name | Student_number | Class | Major |
|---|---|---|---|

**COURSE**

| Course_name | Course_number | Credit_hours | Department |
|---|---|---|---|

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---|---|

**SECTION**

| Section_identifier | Course_number | Semester | Year | Instructor |
|---|---|---|---|---|

**GRADE_REPORT**

| Student_number | Section_identifier | Grade |
|---|---|---|

# Integrity Constraints

- Entity integrity: no NULL in primary keys
- Referential integrity: foreign keys must match primary keys

**STUDENT**

| Name | Student_number | Class | Major |
|---|---|---|---|

**COURSE**

| Course_name | Course_number | Credit_hours | Department |
|---|---|---|---|

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---|---|

**SECTION**

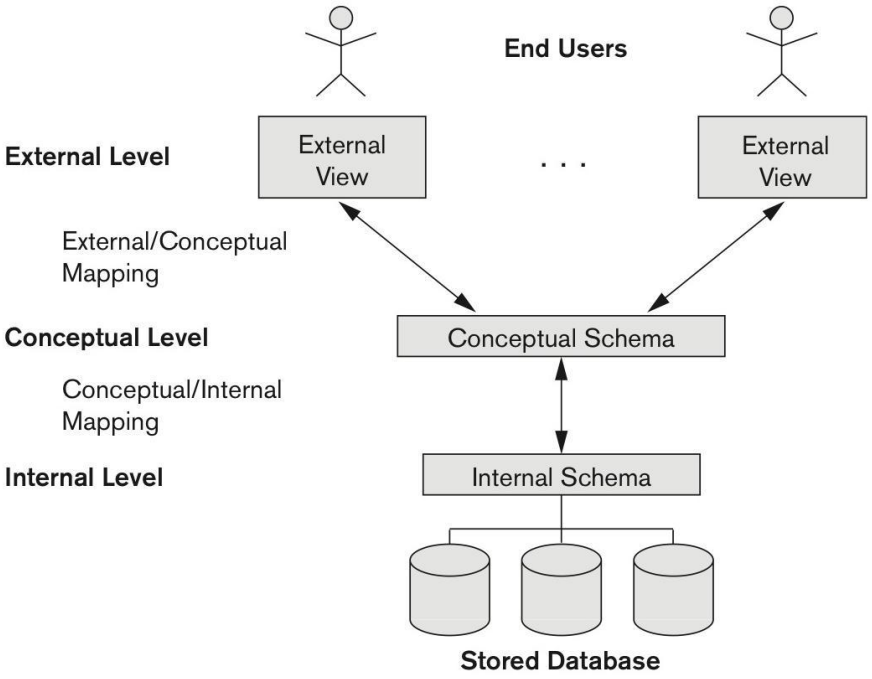
| Section_identifier | Course_number | Semester | Year | Instructor |
|---|---|---|---|---|

**GRADE_REPORT**

| Student_number | Section_identifier | Grade |
|---|---|---|

# Schemas vs Instances

- Schema = description (changes rarely)
- Instance = data state (changes frequently)
- Schema = intension, Instance = extension

# Data Independence & Three-Schema Architecture

- Logical: change conceptual schema without affecting views
- Physical: change internal storage without affecting schema